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DOCUMENT 2 — METAL HYDRIDE STORAGE

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Component: Metal Hydride Hydrogen Storage
Purpose:
Store hydrogen thermochemically for seasonal storage applications.

Inputs:
- Hydrogen mass flow mH2_dot [kg/s]
- Absorption temperature T_abs_K [K]

Outputs:
- Stored hydrogen
- Desorption heat

Key Variables:
T_abs_K : absorption temperature [K]
T_des_K : desorption temperature [K]
Peq_Pa : equilibrium pressure [Pa]

Equations:

$$\ln(Peq\_Pa) = A - B / T\_abs\_K$$

$$Peq\_Pa = \exp(A - B / T\_abs\_K)$$

$$T\_abs\_K = T\_abs\_K$$

$$T\_des\_K = T\_des\_K$$

$$Peq\_Pa = Peq\_Pa$$

Assumptions:
- Equilibrium thermodynamics
- Lumped thermal behaviour